

Formal Verification Supplement: Exact Hulls and Semidefinite Lifts

1 Verified exact hulls and finite semidefinite lifts

For $r \geq 2$, let

$$S = \{(u, v) \in \mathbb{R}^r \times \mathbb{R}^r : \|u\|^2 < 1, \|v\|^2 < 1, u^\top v > 1/2\},$$

and let T denote the system obtained by replacing all three inequalities by weak inequalities. Write $p = 1 - \|u\|^2$, $q = 1 - \|v\|^2$, and $d = u^\top v$. This Lean package proves

$$\begin{aligned} \text{conv } S &= \{(u, v) : p > 0, q > 0, d + \sqrt{pq} > 1/2\}, \\ \text{cl}(\text{conv } S) &= \text{conv } T = \{(u, v) : p \geq 0, q \geq 0, d + \sqrt{pq} \geq 1/2\}. \end{aligned}$$

Every equality includes $r = 2$. The proof of sufficiency is a direct two-point construction, so it does not depend on an unformalized general good-aggregation hull theorem.

A two-point proof in every required dimension. Suppose the strict scalar conditions hold. Put $a = \sqrt{p}$ and $b = \sqrt{q}$. Choose a unit vector e perpendicular to $bu - av$; this requires only $r \geq 2$. Orthogonality implies

$$\frac{u^\top e}{a} = \frac{v^\top e}{b} = h.$$

Choose

$$\max \left\{ 0, \frac{1/2 - d}{ab} \right\} < k < 1, \quad t_\pm = -h \pm \sqrt{h^2 + k}.$$

The two roots have opposite signs and satisfy $2ht_\pm + t_\pm^2 = k$. For $x(t) = (u + tae, v + tbe)$ and $t \in \{t_-, t_+\}$, direct expansion gives

$$\|u + tae\|^2 = \|u\|^2 + pk, \quad \|v + tbe\|^2 = \|v\|^2 + qk, \quad (u + tae)^\top (v + tbe) = d + abk.$$

Thus both $x(t_-)$ and $x(t_+)$ lie in S . The positive weights $t_+/(t_+ - t_-)$ and $-t_-/(t_+ - t_-)$ sum to one and recover (u, v) . This proves hull sufficiency, including the four-variable case.

Necessity follows from the already verified validity of every good-cone aggregation on $\text{conv } S$. The coordinate multipliers give $p, q > 0$; the multiplier $(q, p, 2\sqrt{pq})$ gives $d + \sqrt{pq} > 1/2$. The package also proves the weak cone-test implication, with explicit separating multipliers when $p = 0$ or $q = 0$. It does not infer the weak claim by replacing strict signs in the strict proof.

Actual affine matrix lifts. Define the symmetric matrix

$$L(u, v, \sigma) = \begin{pmatrix} 1 & \sigma & u^\top \\ \sigma & 1 & v^\top \\ u & v & I_r \end{pmatrix}.$$

The formalization proves both the PSD and positive definite Schur-complement equivalences for this matrix and eliminates σ . Consequently,

$$\begin{aligned} (u, v) \in \text{conv } S &\iff \exists \sigma > 1/2 : L(u, v, \sigma) \succ 0, \\ (u, v) \in \text{cl}(\text{conv } S) &\iff \exists \sigma \geq 1/2 : L(u, v, \sigma) \succeq 0. \end{aligned}$$

The latter representation also describes $\text{conv } T$. Singular Schur complements and both signs of $\sigma - d$ are included.

Closure and the original weak system. For a point in the weak scalar region and $0 \leq s < 1$, the scalar margin of (su, sv) above $1/2$ is at least $(1 - s^2)/2$. This follows from concavity of the square root of the two-by-two determinant. Thus radial contraction approximates every weak-region point by strict-hull points. Continuity gives the reverse closure inclusion.

To prove equality with $\text{conv } T$, the proof uses compactness of T and of the image of $[0, 1] \times T \times T$ under $(\alpha, y, z) \mapsto \alpha y + (1 - \alpha)z$. The two-point construction places the strict hull inside this compact image, hence its closure lies there as well. The image is contained in $\text{conv } T$; convexity of the weak scalar region gives the other inclusion.

Finally, the package identifies $\text{conv } S$ with the intersection of all strict good aggregations, and $\text{cl}(\text{conv } S)$ with the intersection of all weak good aggregations. Goodness retains its source definition using the actual homogeneous eigenvalue count and ordinary-hull validity. These equalities supply the exact descriptions for which the preceding package proved that no countable strict good description and no finite weak good description can suffice.

Scope. The frozen claims and declaration coverage are in `supplement/lean/audits/30-infinite-aggregation-hull/`.

These sources extend `Formal.InfiniteAggregation` without changing the earlier certified modules. This package does not verify a numerical SDP solver, quantitative approximation, arbitrary-quadratic impossibility, or a general aggregation hull theorem. The accuracy bounds have a separate verification package.